

3.

g. zv. funkc., $g(0)=0$

a) dokaži: $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$

$$f(x) = (1+x)^{\frac{1}{x}}; \quad f(0) = e$$

$$\forall \varepsilon > 0 \exists \delta > 0 : x \in (-\delta, \delta) \setminus \{0\} : |f(x) - e| < \varepsilon$$

$$\lim_{h \rightarrow \infty} f\left(\frac{1}{h}\right) = \lim_{h \rightarrow \infty} \left(1 + \frac{1}{h}\right)^{\frac{1}{1/h}} = e$$

že vemo

$$x \in (0, 1):$$

$$n_x \in \mathbb{N}:$$

$$\frac{1}{n_x+1} < x < \frac{1}{n_x} \quad (\Leftrightarrow) \quad n_x < \frac{1}{x} < n_x+1$$

$$\wedge \frac{1}{x} \left(\begin{array}{l} \text{najdemo tak } n_x, \text{ da velja} \end{array} \right.$$

$$\left(1 + \frac{1}{n_x+1}\right)^{\frac{1}{x}} \leq \underbrace{\left(1+x\right)^{\frac{1}{x}}} \leq \underbrace{\left(1 + \frac{1}{n_x}\right)^{\frac{1}{x}}} < \underbrace{\left(1 + \frac{1}{n_x}\right)^{n_x+1}}$$

$\downarrow$ $\downarrow$
 e e

po prim. krit. je tudi $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$

$$\exists N \in \mathbb{N}; \quad \forall n \geq N:$$

$$e - \varepsilon < \left(1 + \frac{1}{n+1}\right)^n < \left(1 + \frac{1}{n}\right)^{n+1} < e + \varepsilon$$

$$\text{torej } \varepsilon = 1 \quad \Rightarrow \quad 0 < x < \varepsilon \quad \Rightarrow \quad x > 1/n$$

$$e - \xi < \left(1 + \frac{1}{h_x + 1}\right)^{h_x} \leq (1+x)^{\frac{1}{x}} \leq \left(1 + \frac{1}{h_x}\right)^{h_x + 1} < e + \xi$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \left(1 + \frac{1}{x}\right)^x = e$$

b) Dokaži: $\lim_{x \rightarrow 0} \left(1 + g(x)\right)^{\frac{1}{g(x)}} = e$

// $\leftarrow f \circ g$ zvezna

$$\lim_{x \rightarrow 0} f(g(x)) = f\left(\lim_{x \rightarrow 0} g(x)\right) = f(0) = e$$

1.) g zv., $g(0) = 0$

$$\lim_{x \rightarrow 0} \left(1 + g(x)\right)^{\frac{1}{g(x)}} = e$$

2.) g zv., $g(0) = 0$, $\lim_{x \rightarrow 0} h(x) = \pm \infty$

$$\lim_{x \rightarrow 0} \left(1 + g(x)\right)^{h(x)} = e^{\lim_{x \rightarrow 0} (g(x)h(x))}$$

4.

a) $\lim_{x \rightarrow 0} \left(1 + \sin x\right)^{\frac{1}{x}} = e^{\lim_{x \rightarrow 0} \frac{\sin x}{x}} = e$

b) $\lim_{x \rightarrow 0} (\cos x)^{\frac{1}{\sin^2 x}} = \lim_{x \rightarrow 0} (1 - \sin^2 x)^{\frac{1}{2\sin^2 x}} = e^{\lim_{x \rightarrow 0} \frac{-\sin^2 x}{2\sin^2 x}} = e^{-\frac{1}{2}}$

c) $\lim_{x \rightarrow 0} (\sin x + \cos x)^{\frac{1}{\sin x}} = \lim_{x \rightarrow 0} \underbrace{\left(1 + \sin x + \cos x - 1\right)}_{g(x) [g(0)=0]}^{\frac{1}{\sin x}} = e^{\lim_{x \rightarrow 0} \frac{(\sin x + \cos x - 1)\cos x}{\sin x}}$

$$= e^{1 + \lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x + 1)}{\sin x (\cos x + 1)}} = e^{1 + \lim_{x \rightarrow 0} \frac{\sin x}{\cos x + 1}} = e^{1+0} = e$$

5. D_f , $\lim_{x \rightarrow a} \underbrace{u_n}_{\text{robu } D_f}$

a) ∂D_f

$$f(x) = \frac{1}{1+e^{\frac{1}{x}}}$$

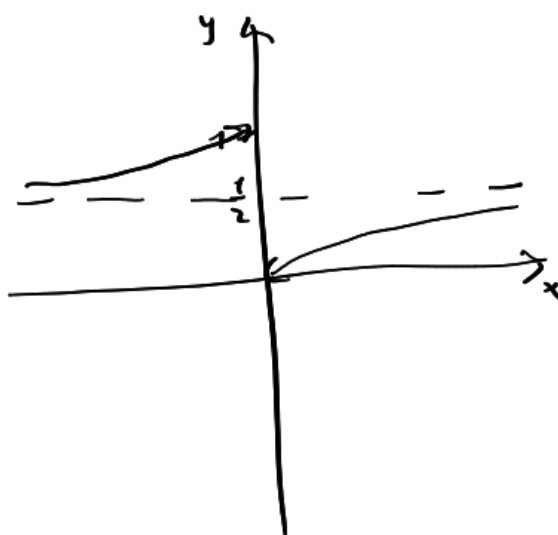
$$D_f = \mathbb{R} \setminus \{0\}$$

$$\lim_{x \rightarrow 0^+} \frac{1}{1+e^{\frac{1}{x}}} = 0$$

$$\lim_{x \rightarrow 0^-} \frac{1}{1+e^{\frac{1}{x}}} = 1$$

$$\lim_{x \rightarrow \infty} \frac{1}{1+e^{\frac{1}{x}}} = \frac{1}{2}$$

$$\lim_{x \rightarrow -\infty} \frac{1}{1+e^{\frac{1}{x}}} = \frac{1}{2}$$



b) $f(x) = (x+1) \operatorname{arctg}\left(\frac{1}{1-x^2}\right)$

$$D_f = \mathbb{R} \setminus \{1, -1\}$$

$$\frac{1}{1-x^2}$$



$$\lim_{x \rightarrow 1^+} (x+1) \operatorname{arctg}\left(\frac{1}{1-x^2}\right) = -\pi$$

$$\lim_{x \rightarrow 1^-} (x+1) \operatorname{arctg}\left(\frac{1}{1-x^2}\right) = \pi$$

$$\lim_{x \rightarrow 1^+} f(x) = 0$$

$$\lim_{x \rightarrow -1^-} f(x) = 0$$

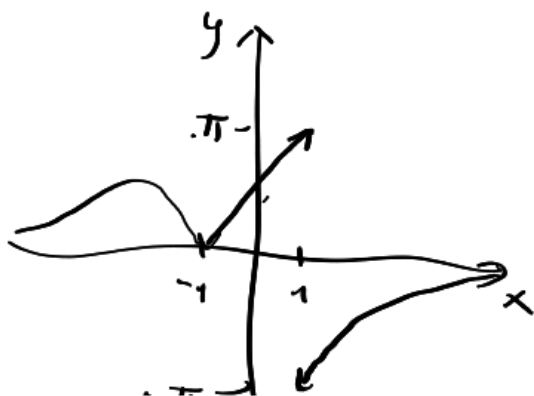
$$\lim_{x \rightarrow 1^-} \frac{\operatorname{arctg} x}{x} = \lim_{t \rightarrow \pi/2^-} \frac{t \cdot \cos t}{\sin t} = 1$$

$x = \operatorname{tg} t$
 $t = \operatorname{arctg} x$



$$\lim_{x \rightarrow \infty} (x+1) \operatorname{arctg}\left(\frac{1}{1-x^2}\right) = x+1 \cdot \frac{\operatorname{arctg}\left(\frac{1}{1-x^2}\right)}{\frac{1}{1-x^2}} \cdot \frac{1}{1-x^2} = 0$$

$$\lim_{x \rightarrow -\infty} f(x) = 0$$



6. Bisekcija, na 1 decimalke nat.

$$x-1 = \frac{1}{2} \sin x$$

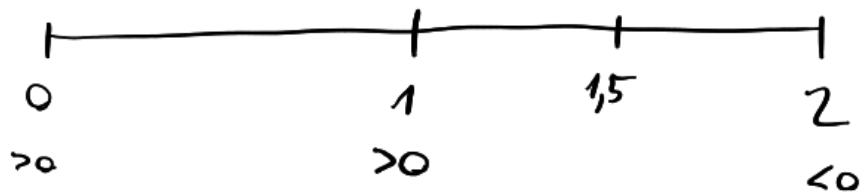
$$f(x) = \frac{1}{2} \sin x - x + 1$$

$$f(0) = 1$$

$$f(2) = -0,545 < 0$$

$$f(1) = 0,42 > 0$$

$$f(1,5) = -0,001$$

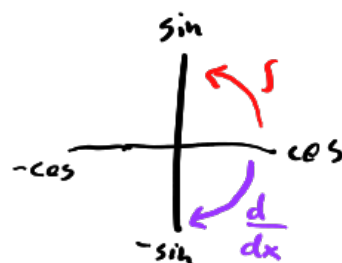


$$x = 1,5 \text{ (na 1 dec. nat.)}$$

ODVODI

odvod inverza:

$$(f^{-1})' = \frac{1}{f'(f^{-1}(x))}$$



1. Izračunaj odvode:

a) arcsinh

$$(\sinh x)' = \left(\frac{e^x - e^{-x}}{2} \right)' = \frac{1}{2} (e^x + e^{-x}) = \cosh x$$

$$\begin{aligned} (\text{arcsinh } x)' &= \frac{1}{\cosh(\text{arcsinh } x)} = \\ &= \frac{1}{\sqrt{1 + \sinh^2(\text{arcsinh } x)}} = \frac{1}{\sqrt{1 + x^2}} \end{aligned}$$

$$\begin{aligned} \cosh^2 x - \sinh^2 x &= 1 \\ \cosh x &= \sqrt{1 + \sinh^2 x} \end{aligned}$$

b) arch

$$(\cosh x)' = \left(\frac{e^x + e^{-x}}{2} \right)' = \frac{e^x - e^{-x}}{2} = \sinh x$$

$$(\text{arch } x)' = \frac{1}{\sinh(\text{arch } x)} = \frac{1}{\sqrt{\cosh^2(\text{arch } x) - 1}} = \frac{1}{\sqrt{x^2 - 1}}$$

c) $\operatorname{arcth} x$

$$\operatorname{th} x = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^x(1 - e^{-2x})}{e^x(1 + e^{-2x})} = \frac{1 - e^{-2x}}{1 + e^{-2x}} \quad \operatorname{ch}^2 x - \operatorname{sh}^2 x = 1$$

$$(\operatorname{th} x)' = \left(\frac{\operatorname{sh} x}{\operatorname{ch} x} \right)' = \frac{(\operatorname{sh} x)' \operatorname{ch} x - \operatorname{sh} x (\operatorname{ch} x)'}{(\operatorname{ch} x)^2} = \frac{1}{\operatorname{ch}^2 x} \quad \frac{1}{\operatorname{ch}^2 x} = 1 - \operatorname{th}^2 x$$

$$(\operatorname{arcth} x)' = \frac{1 \cdot \operatorname{ch}^2(\operatorname{arcth} x)}{1} = \operatorname{ch}^2(\operatorname{arcth} x) = \frac{1}{1 - \operatorname{th}^2(\operatorname{arcth} x)} = \frac{1}{1 - x^2}$$

2. Izračunaj odvode

a) $f(x) = x^x = e^{x \ln x}$

$$f'(x) = e^{x \ln x} \cdot \left(\ln x + x \cdot \frac{1}{x} \right) = x^x (\ln x + 1)$$

b) $f(x) = (1+x)^{\frac{1}{x}} = e^{\frac{1}{x} \ln(1+x)}$

$$\begin{aligned} f'(x) &= (1+x)^{\frac{1}{x}} \left(-x^{-2} \ln(1+x) + \frac{1}{x} \cdot \frac{1}{x+1} \cdot 1 \right) = \\ &= (1+x)^{\frac{1}{x}} \left(\frac{1}{x(x+1)} - \frac{\ln(1+x)}{x^2} \right) = (1+x)^{\frac{1}{x}} \left(\frac{x - (x+1) \ln(x+1)}{x^2(x+1)} \right) = \\ &= (x+1)^{\frac{1}{x}} \left(\frac{x - \ln(x+1)^{x+1}}{x^2(x+1)} \right) \end{aligned}$$

Lopital

3.

a) $\lim_{x \rightarrow 0^+} \frac{e^x - 1}{\ln(x+1)} = \lim_{x \rightarrow 0^+} \frac{e^x}{\frac{1}{x+1}} = 1$

b) $\lim_{x \rightarrow 0^+} x^h \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{x^{-h}} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-h x^{-h-1}} = \lim_{x \rightarrow 0^+} x^{-1} \cdot x^{h+1} \cdot \frac{1}{-h} = \lim_{x \rightarrow 0^+} -\frac{x^h}{h} = 0$

